Fluid Dynamics II

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1 Introduction

The Part IB Fluid Dynamics course mainly covers inviscid flow and irrotational flow including potential flow. The course covered an introduction to viscous flow for the very simple case where $\mathbf{u} = (u(y,t), 0)$ in 2D parallel flow.

This course is about the effects of viscosity and answers the following very important questions

(i) How does viscosity affect the equations for a general flow?

(ii) What happens if viscosity dominates over the inertia of the fluid?

(iii) What happens if the flow is nearly parallel?

(iv) What happens if viscosity is small compared to inertia?

(v) What about stability?

2 Equations for a Newtonian Viscous Flow

2.1 Dynamic viscosity

Recall from Part IB Fluid Dynamics that we have Couette flow, in which a plate at rest and a plate moving with velocity U are parallel, with a gap h between them filled with a viscous fluid. For a simple fluid, we find that the force required to move the top plate F is proportional to $\frac{UA}{h}$. Hence the shear stress is

$$\tau_s = \frac{F}{A} = \mu \frac{U}{h}$$

where μ is the dynamic viscosity. The shear stress is due to the relative motion between each layer of fluid and its neighbours above and below.

2.2 Rate of strain tensor

Consider the velocity near a fixed point $\mathbf{0}$ *without loss of generality*. Taylor expand $\mathbf{u}$, so we have the multivariable Taylor expansion

$$u_i(\mathbf{x}) = u_i(0) + x_j \left.\frac{\partial u_i}{\partial x_j}\right|_{\mathbf{0}} + \frac{1}{2} x_j x_k \left.\frac{\partial^2 u_i}{\partial x_j \partial x_k}\right|_{\mathbf{0}} + \cdots$$

We drop the quadratic terms and decompose $\frac{\partial u_i}{\partial x_j}$ into symmetric and antisymmetric parts, so

$$\begin{aligned}\frac{\partial u_i}{\partial x_j} &= \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) \\ &= e_{ij} + \frac{1}{2}\varepsilon_{jik}\omega_k\end{aligned}$$

where e_{ij} is the *rate of strain tensor* and $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ is the vorticity. Hence we have that

$$\mathbf{u}(\mathbf{x}) \approx \mathbf{u}(\mathbf{0}) + \mathbf{e} \cdot \mathbf{x} + \frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}.$$

We have that $e_{ij} = e_{ji}$ by definition so $\mathbf{e}$ is real symmetric and hence has real orthogonal eigenvectors, which are the principal axes of strain. For an incompressible flow we have that $e_{ii} = \nabla \cdot \mathbf{u} = 0$ so the eigenvalues sum to 0. The eigenvalues are called the principal rates of strain.

2.3 The stress tensor

Recall in mechanics we consider two types of forces:

(i) Body forces $\mathbf{f}$ that act per unit volume, such as gravity;

(ii) Surface forces $\boldsymbol{\tau}$ that act per unit area, for example, pressure and viscous shear stress

Let $\boldsymbol{\tau}(\mathbf{n}; \mathbf{x}, t)\delta S$ denote the force exerted by the positive side on the negative side of the small area element with $\mathbf{n}$ pointing from the negative side to the positive side.

For example, we have that $\boldsymbol{\tau}(\mathbf{n}) = -p\mathbf{n}$ for an inviscid fluid.

Now let $\mathbf{n}\delta S$ be the sloping face of a small tetrahedron of size ℓ whose three other faces are aligned with the coordinate planes. The force acting on the sloped face is $\boldsymbol{\tau}(\mathbf{n})\delta S$. The forces on the other faces are $\boldsymbol{\tau}(-\mathbf{1})\delta S_1, \boldsymbol{\tau}(-\mathbf{2})\delta S_2, \boldsymbol{\tau}(-\mathbf{3})\delta S_3$. Considering the force balance, the body force and inertia are $O(\ell^3)$ and the surface forces $\boldsymbol{\tau}$ are $O(\ell^2)$, so their ratio is $O(\ell) \to 0$ as $\ell \to 0$. So the surface forces must balance themselves for small ℓ.

Note that pressure and viscous stresses balance inertia and gravity on the characteristic length scale, so small ℓ means that ℓ is small compared to the characteristic length scale L.

Since the forces balance, we have that

$$\boldsymbol{\tau}(\mathbf{n})\delta S + \boldsymbol{\tau}(-\mathbf{1})\delta S_1 + \boldsymbol{\tau}(-\mathbf{2})\delta S_2 + \boldsymbol{\tau}(-\mathbf{3})\delta S_3 = 0.$$

Now

$$\delta S_i = \mathbf{i} \cdot (\mathbf{n}\delta S) = n_i \delta S$$

and $\boldsymbol{\tau}(-\mathbf{1}) = -\boldsymbol{\tau}(\mathbf{1})$ by Newton's Third Law. Substituting these in, we get

$$\boldsymbol{\tau}(\mathbf{n}) = \boldsymbol{\tau}(\mathbf{1})n_1 + \boldsymbol{\tau}(\mathbf{2})n_2 + \boldsymbol{\tau}(\mathbf{3})n_3$$

which shows that $\boldsymbol{\tau}$ is linear in $\mathbf{n}$. So we have that

$$\tau_i(\mathbf{n}) = \sigma_{ij} n_j$$

where $\sigma_{ij} = \tau_i(\mathbf{j})$.

Definition. (Stress tensor) We call $\sigma(\mathbf{x}, t)$, as constructed above, the *stress tensor*.

The key result is that the stress $\boldsymbol{\tau}$ exerted on a small area element (by the outside on the inside) is given by $\boldsymbol{\tau}(\mathbf{n}) = \sigma \cdot \mathbf{n}$.

For the inviscid case $\sigma = -p\,\mathbf{I}$, i.e. $\sigma_{ij} = -p\delta_{ij}$, where $\mathbf{I}$ is the identity tensor.

We will now show that σ is a symmetric tensor.

Consider the angular momentum balance on a small ($\ell \ll L$) arbitrary volume V about a fixed point in V, which we call $\mathbf{0}$.

The moment of the surface forces is

$$\int_{\partial V} \varepsilon_{ijk} x_j \sigma_{km} n_m \mathrm{d}S$$

which is $O(\ell^3)$. Similarly, with ρ the density, the moment of the body forces and inertia is

$$\int_V \mathbf{x} \times \left(\mathbf{f} - \rho \frac{D\mathbf{u}}{Dt} \right) \mathrm{d}V$$

which is also $O(\ell^4)$.

For small ℓ, the moments of the surface forces must balance, hence

$$0 = \int_V \varepsilon_{ijk} \frac{\partial}{\partial x_m}(x_j \sigma_{km}) \mathrm{d}V = \int_V \varepsilon_{ijk} \left(\sigma_{kj} + x_j \frac{\partial \sigma_{km}}{\partial x_m} \right) \mathrm{d}V.$$

Since x_j is $O(\ell)$, we can take $\ell \to 0$ to get rid of the second term, so the integral is approximately

$$\int_V \varepsilon_{ijk} \sigma_{kj} \mathrm{d}V.$$

Since V is arbitrary, $\varepsilon_{ijk}\sigma_{kj} = 0$, hence $\sigma_{ij} = \sigma_{ji}$.

2.4 Constitutive Equation for a Newtonian Fluid

We divide the stress tensor into two terms:

(i) An isotropic pressure term $-p\delta_{ij}$, with $p = -\frac{1}{3}\sigma_{kk}$.

(ii) A deviatoric stress $\sigma_{ij}^{\mathrm{dev}}$ which is the deviation from isotropic and so is traceless: $\sigma_{kk}^{\mathrm{dev}} = 0$.

So we have that

$$\sigma_{ij} = -p\delta_{ij} + \sigma_{ij}^{\mathrm{dev}}.$$

The deviatoric stress is due to frictional interactions with neighbouring fluid particles sliding past with slightly different velocities. For fluid at rest, we must have a static equilibrium where $\nabla p = \mathbf{f}$. Relative motion in the neighbourhood of a point is given by the velocity gradient, so we expect that

$$\sigma^{\mathrm{dev}} = \mathcal{F}[\nabla \mathbf{u}]$$

and $\mathcal{F}(\mathbf{0}) = \mathbf{0}$. A Newtonian fluid is defined by two properties.

(i) σ^{dev} is linear in the local, instantaneous value of $\nabla \mathbf{u}$, hence we have that

$$\sigma_{ij}^{\mathrm{dev}} = A_{ijk\ell} \frac{\partial u_k}{\partial x_\ell}$$

for some tensor $A_{ijk\ell}$.

Local instananeous is good for simple (Newtonian) fluids for which $\ell_{\mathrm{mol}} \sim 10^{-9}$ m and $t_{\mathrm{mol}} \sim 10^{-12}$ s. This can go wrong since we can have action at a distance such as a flow which contains a suspension of fibres. The fluid might also have memory due to deformation elasticity such as a solution contain long-chain polymers.

(ii) The fluid is locally isotropic. This is not true for example in a liquid crystal. So $A_{ijk\ell}$ must be an isotropic tensor, hence

$$A_{ijk\ell} = \mu' \delta_{ij}\delta_{k\ell} + \mu'' \delta_{ik}\delta_{j\ell} + \mu''' \delta_{i\ell}\delta_{jk}.$$

Together this gives us that

$$\sigma_{ij}^{\text{dev}} = \mu' \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu'' \frac{\partial u_j}{\partial x_i} + \mu''' \frac{\partial u_i}{\partial x_j}$$

and luckily for us the first term is zero since the flow is assumed to be incompressible. Now since σ is symmetric we have that $\mu'' = \mu'''$, so let's drop the primes and call them both μ.

This gives us

$$\begin{aligned}\sigma_{ij} &= -p\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \\ &= -p\delta_{ij} + 2\mu e_{ij}.\end{aligned}$$

This is called the constitutive relationship for an incompressible Newtonian viscous fluid.

Remark. We'll make some remarks on this relationship.

(i) No dependence on the vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, only depends on the strain-stress rate $e = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$, since localy solid-body rotation $\frac{1}{2}\boldsymbol{\omega} \times \mathbf{x}$ doesn't involve sliding or friction.

(ii) Many common fluids (air, water, oil, syrup) match our assumptions for getting to these equations up to a very close approximation.

(iii) μ here is the same as in IB FLuid Dynamics. Since if we take a flow $\mathbf{u} = (u(y), 0, 0)$ we find that $\sigma_{xy} = \mu \frac{\partial u}{\partial y}$ as we did last year.

2.5 The Momentum Equation

2.5.1 The Cauchy Momentum Equation

The momentum in an arbitrary fixed volume V can change with time due to

(i) Momentum being carried across the boundary ∂V;

(ii) Applying body forces acting on V;

(iii) Applying surface forces on ∂V.

So we have that

$$\frac{d}{dt}\int_V \rho u_i \mathbf{d}V = -\int_{\partial V} \rho u_i (\mathbf{u} \cdot \mathbf{n}) \mathrm{d}S + \int_V f_i \mathrm{d}V + \int_{\partial V} \sigma_{ij} n_j \mathrm{d}S.$$

We'll use that fact that V is fixed and apply the divergence theorem, so

$$\int_V \frac{\partial}{\partial t}(\rho u_i)\mathrm{d}V + \int_V \frac{\partial}{\partial x_j}(\rho u_i u_j)\mathrm{d}V = \int_V \left(f_i + \frac{\partial}{\partial x_j}\sigma_{ij}\right)\mathrm{d}V.$$

Now since V is arbitrary we can drop the integral, so we get

$$u_i\left(\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j)\right) + \rho\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) = f_i + \frac{\partial \sigma_{ij}}{\partial x_j}.$$

Notice that the first bracketed term is the left-hand side of the mass conservation equation, so it is zero. This gives us the Cauchy momentum equation,

$$\rho \frac{D\mathbf{u}}{Dt} = \rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u}\right) = \mathbf{f} + \nabla \cdot \sigma.$$

2.5.2 The Navier-Stokes Equation

For a Newtonian fluid $\sigma_{ij} = -p\delta_{ij} + 2\mu e_{ij}$. We assume that μ is uniform in the fluid, so we get that

$$\frac{\partial}{\partial x_j}\sigma_{ij} = -\frac{\partial p}{\partial x_j}\delta_{ij} + \mu\left(\frac{\partial^2 u_i}{\partial x_j \partial x_j} + \frac{\partial^2 u_j}{\partial x_j \partial x_i}\right).$$

And we can use the fact that the fluid is incompressible, to get the *Navier-Stokes equation*

$$\rho\frac{D\mathbf{u}}{Dt} = \mathbf{f} - \nabla p + \mu\nabla^2\mathbf{u}.$$

2.5.3 Boundary conditions

For a material boundary with normal $\mathbf{n}$ we have that

(i) Mass conservations gives that $\mathbf{u}\cdot\mathbf{n}$ is continuous;

(ii) $\nabla\mathbf{u}$ is finite because we need σ to be finite, so $\mathbf{u} - (\mathbf{u}\cdot\mathbf{n})\mathbf{n}$ is continuous for a viscous fluid. This is the no slip condition. These together mean that $\mathbf{u}$ is continuous.

(iii) From Newton's Third law we get that $\sigma\cdot\mathbf{n}$ is continuous since the stresses must balance on the boundary. i.e the body stress on the fluid is the negative of the fluid stress on the solid.

2.6 Energy Equation and Viscous Dissipation

Obtain a mechanical energy equation by dotting the Cauchy momentum equation with $\mathbf{u}$ so we get

$$u_i\rho\left(\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j}\right) = u_i f_i + u_i\frac{\partial \sigma_{ij}}{\partial x_j}$$

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho u^2\right) - \frac{1}{2}u^2\frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x_j}\left(\frac{1}{2}\rho u^2 u_j\right) - \frac{1}{2}u^2\frac{\partial}{\partial x_j}(\rho u_j) = u_i f_i + \frac{\partial}{\partial x_j}(u_i\sigma_{ij}) - \sigma_{ij}\frac{\partial u_i}{\partial x_j}.$$

Now we integrate over a fixed volume V to get the following equation.

Definition. (Energy Equation)

$$\frac{d}{dt}\int_V \frac{1}{2}\rho u^2 \,\mathrm{d}V + \int_{\partial V}\frac{1}{2}\rho u^2(\mathbf{u}\cdot\mathbf{n})\,\mathrm{d}S = \int_V \mathbf{u}\cdot\mathbf{f}\,\mathrm{d}V + \int_{\partial V}\mathbf{u}\cdot\sigma\mathbf{n}\,\mathrm{d}S - \int_V \sigma_{ij}e_{ij}\,\mathrm{d}V.$$

Now we can explain each term in the equation in the order as written.

(i) The rate of change of kinetic energy inside V.

(ii) Kinetic energy being carried across the boundary ∂V.

(iii) The rate of doing work by body forces inside V.

(iv) The rate of doing work by surface forces $\boldsymbol{\tau} = \sigma\cdot\mathbf{n}$ on ∂V.

(v) The rate of doing work against the internal stresses due to deformation.

For incompressible Newtonian fluids $\sigma_{ij} = -p\delta_{ij} + 2\mu e_{ij}$ so we have that

$$\sigma_{ij}e_{ij} = -pe_{ii} + 2\mu e_{ij}e_{ij}$$

and e_{ii} vanishes since the flow is incompressible.

Hence in the case for an incompressible fluid, the fifth term gives the rate of loss of mechanical energy (dissipation) due to viscosity and defomration. The energy lost if converted to heat and is a source term in the temperature equation.

Definition. (Localy viscous dissipation rate) We define Φ as the *viscous dissipation rate* given as

$$\Phi = 2\mu e_{ij}e_{ij}.$$

2.7 Nondimensionalisation

Suppose a flow has a characteristic length scale L and velocity scale U and varies on a timescale T. Assume no body forces so we get that

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0$$

where $\mathbf{u} = \mathbf{u}(\mathbf{x}, t, \rho, \mu, U, L, T, \text{shape}, \dots)$. Define a dimensionless velocity $\mathbf{u}^* = \frac{\mathbf{u}}{U}$, position $\mathbf{x}^* = \frac{\mathbf{x}}{L}$, time $t^* = \frac{t}{T}$, pressure $p^* = \frac{p}{\mu U/L}$ (an alternative inertial scaling sets $p^{**} = \frac{p}{\rho U^2}$).

And substitute these into the Navier-Stokes equation. This gives that

$$\rho\left(\frac{U}{T}\frac{\partial u^*}{\partial t} + \frac{U^2}{L}\mathbf{u}^*\cdot\nabla^*\mathbf{u}^*\right) = -\frac{\mu U}{L^2}\nabla^* p^* + \frac{\mu U}{L^2}\nabla^{*2}\mathbf{u}^*.$$

So collecting constants, we get that

$$\mathrm{Re}\left(\mathrm{St}\frac{\partial u^*}{\partial t^*} + (\mathbf{u}^*\cdot\nabla)\mathbf{u}^*\right) = -\nabla^* p^* + \nabla^{*2}\mathbf{u}^*$$

where we have the following defintions

Definition. (Reynolds number) For a flow with length scale L, velocity scale U, density ρ, and dynamic velocity μ we have

$$\mathrm{Re} = \frac{\rho U L}{\mu} = \frac{UL}{\nu}$$

where we are defining

$$\nu = \frac{\mu}{\rho}$$

to be the *kinematic viscosity.*

We also have $\mathrm{St} = \frac{L}{UT}$ as the *Stroughal number* although this comes up much less.

These are dimensionless quantities which we can use to characterise properties of the flow.

Now we can have $\mathbf{u}^* = \mathbf{u}^*(\mathbf{x}^*, t, \mathrm{Re}, \mathrm{St}, \mathrm{Shape})$ which is a function of fewer parameters. If the flow is steady then we can set $T = \frac{L}{U}$, called the *advection time*, and $\mathrm{St} = 1$, so the Reynolds number is the only important parameter. If the boundary conditions are oscillatory, then set $T = \frac{1}{\omega}$, and then group

$$\mathrm{Re}_T = \mathrm{Re} \cdot \mathrm{St}$$

is called the *unsteady Reynolds number* or the *Stokes number*.

$$\mathrm{Re}_T = \frac{L^2}{\nu T} = \frac{\omega L^2}{\nu}.$$